

EXPERIMENT 3

SURVEY REPORT ON EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

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Application Topic: Library Book Issue Request Form

1. AIM

To study the emerging trends in low-code application development and understand how low-code platforms can be used to create a Library Book Issue Request Form with minimal hand-written programming. The survey also aims to compare traditional application development with low-code development, identify its benefits and challenges, and compare popular low-code platforms.

2. TRADITIONAL APPLICATION DEVELOPMENT VS EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

Traditional application development requires developers to write and maintain a large amount of source code for the user interface, business logic, database connectivity, validation, security, and deployment. It provides detailed control but generally requires more development time and specialized programming skills.

Low-code application development uses visual interfaces, drag-and-drop components, pre-built controls, reusable workflows, connectors, and configuration instead of requiring developers to write every part of the application manually. It is increasingly used for rapid development of forms, workflows, internal applications, and business automation.

Aspect	Traditional Development	Low-Code Development
Development method	Mainly manual coding	Visual development and configuration
Programming skill	Usually requires strong programming knowledge	Less coding knowledge required
Development time	Generally longer	Generally faster
UI creation	Built using code/frameworks	Drag-and-drop components
Database integration	Often configured through code/APIs	Pre-built connectors and visual configuration
Workflow automation	Implemented using custom code	Can be created using visual workflows
Customization	Very high	High, with platform-specific limitations
Maintenance	Developer-dependent	Many platform features simplify maintenance
Deployment	Requires configuration and deployment steps	Often simplified through platform tools
Typical use	Complex and highly customized	Forms, portals, workflows, business

	systems	apps
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Emerging trends in low-code development include AI-assisted application building, reusable components, workflow automation, API and third-party integrations, cloud-based development, responsive applications, and citizen development. These trends allow organizations to build and modify applications more quickly while reducing the amount of repetitive coding.

3. BENEFITS AND CHALLENGES IN LOW-CODE DEVELOPMENT

Benefits

- **Faster development:** Applications and forms can be created using ready-made components and visual workflows.
- **Reduced coding effort:** Developers can configure many features without writing extensive code.
- **Easy form creation:** Input fields, validation rules, buttons, and workflows can be configured visually.
- **Rapid prototyping:** A working application can be created quickly and improved based on user feedback.
- **Integration:** Many platforms provide connectors for databases, APIs, email, and other services.
- **Automation:** Repetitive business processes can be automated using workflow rules.
- **Collaboration:** Developers and non-developers can participate in application development.
- **Cloud accessibility:** Many low-code platforms provide browser-based development and deployment.

Challenges

- **Platform dependency:** Applications may become dependent on the selected vendor and its platform.
- **Customization limits:** Very complex requirements may require custom code or workarounds.
- **Security concerns:** Applications still need proper authentication, authorization, data protection, and secure configuration.
- **Performance:** Poorly designed applications may not perform well for large workloads.
- **Migration difficulty:** Moving an application from one platform to another may require significant effort.
- **Learning the platform:** Users still need to understand the platform's tools, rules, and workflow features.
- **Licensing cost:** Advanced features and enterprise capabilities may require paid plans.
- **Governance:** Organizations need standards for access control, data management, testing, and application ownership.

4. COMPARISON OF EXISTING LOW-CODE PLATFORMS

The following table compares commonly used low-code application development platforms based on their general capabilities.

Platform	Main Focus	Form/App Development	Workflow	Integrations	Suitable For
Zoho Creator	Business applications	Strong	Strong	Databases, APIs, Zoho services	SMEs and business workflows
Microsoft Power Apps	Business apps	Strong	Strong with Power Automate	Microsoft 365, Dataverse, connectors	Organizations using Microsoft ecosystem
OutSystems	Enterprise applications	Strong	Strong	Enterprise systems and APIs	Large/complex applications
Mendix	Enterprise low-code	Strong	Strong	APIs and enterprise services	Enterprise application development
AppSheet	No-code/low-code apps	Strong	Strong	Google services and data sources	Data-driven business apps

For the Library Book Issue Request Form, a low-code platform can provide fields such as student name, roll number, book title, author, book ID, issue date, return date, and request status. Validation rules can ensure that required information is entered correctly. A workflow can then send the request to the librarian or update the request status after processing.

5. RESULT

Member Name	Register number	Member Type	Email Address
abc def	240741327	Faculty	Rajalakshmi engineering college thandalam, chennai, kanchipuram, abc@gmail.com, 602106, India

Email Address	Rajalakshmi engineering college thandalam, chennai, kanchipuram, abc@gmail.com, 602106, India
Contact Number	+916785323567
Residential Address	Rajalakshmi engineering college thandalam, chennai, kanchipuram, Tamil Nadu, 602106, India
book category	Computer Science and Engineering
Subject Areas Of Interest	deep learning
Number of Copies	2
Purpose Of Borrowing	for reading
Preferred Issue Date	03-Sep-2026
preferred Collection Time	21:30:46
Agree to library Rules	agreed

The survey on emerging trends in low-code application development was completed successfully. Traditional development and low-code development were compared, and the major benefits and challenges of low-code platforms were identified. Popular platforms such as Zoho Creator, Microsoft Power Apps, OutSystems, Mendix, and AppSheet were compared based on their general application, workflow, and integration capabilities.

The Library Book Issue Request Form can be developed efficiently using a low-code platform by combining visual form components, validation rules, database fields, and workflow automation. This demonstrates how low-code development can reduce manual coding effort and support rapid development of practical applications.